

CO513: Advanced Computer Communication Networks - Lab 06

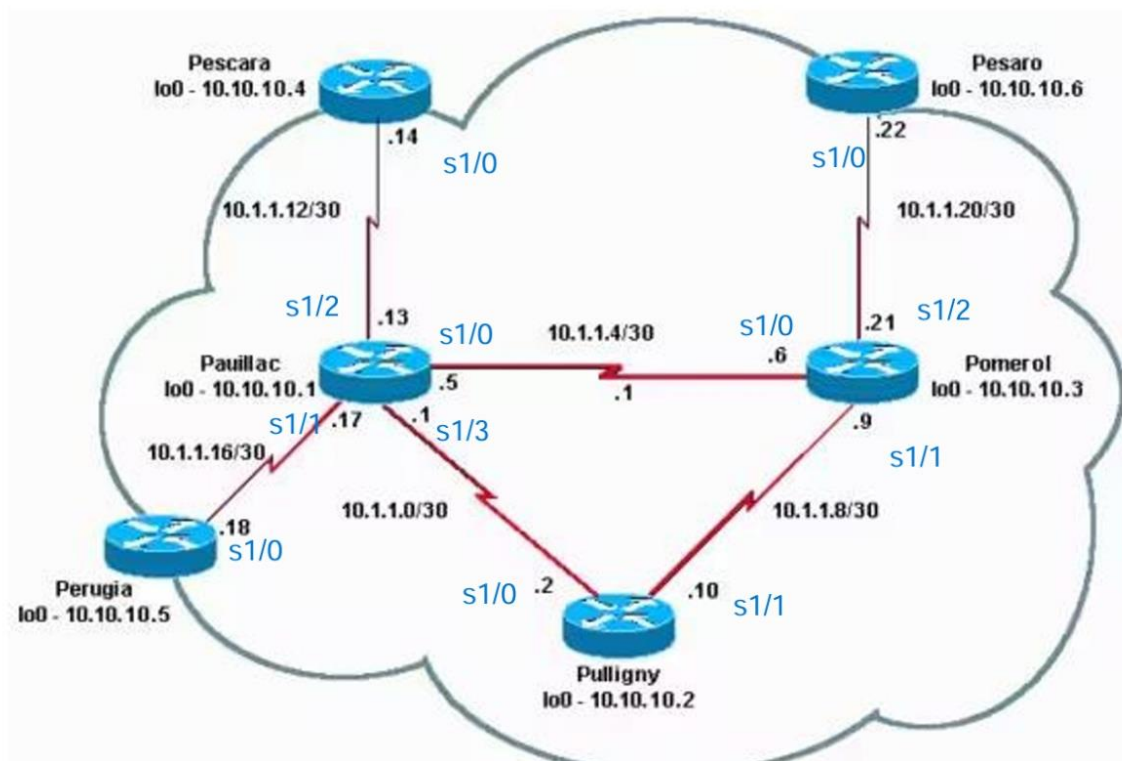
E/19/134, E/19/155, E/19/166, E/19/170,
E/19/205, E/19/210, E/19/309, E/19/349,
E/19/372, E/19/409, E/19/413, E/19/426,
E/19/ 432, E/19/443, E/19/446, E/19/452, E/19/455

04/07/2025

Network Configuration and Setup Summary

We initially attempted to build the network using Aruba Layer 3 switches. However, due to their lack of support for MPLS, we transitioned to a GNS3-based simulation environment.

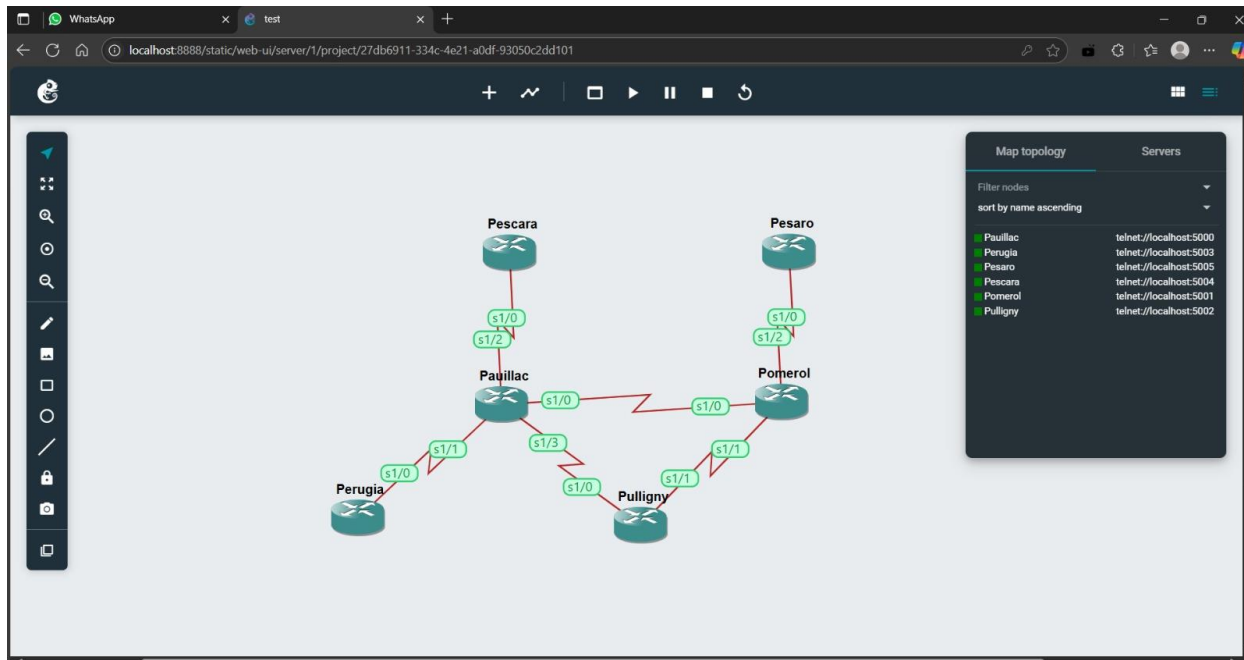
GNS3 was set up on a dedicated department PC, which we accessed remotely (telnet through ssh) to perform all configuration tasks. Cisco 7200 series routers were selected as the core network devices for the simulation, ensuring full MPLS capability.



We set up the physical topology as shown above, with wired connections and assigned IP addresses to router serial interfaces for basic connectivity. OSPF was then configured for dynamic routing, forming the foundation for implementing MPLS, tunneling and VPN in the main lab tasks.

[All complete configurations, including initial ipconfigs and OSPF setup, are provided in the full router configurations attached as Appendices 1 to 8.](#)

Following is a screenshot taken of the initial network setup.



Most of the initial configurations across the six routers are similar; therefore, a **representative example is provided** along with the configuration steps. Significant differences in specific routers are highlighted where relevant. [All complete router configurations, including relevant MPLS VPN settings, Tunneling, and QoS policy configs are provided in Appendices 1 through 8. Each appendix contains the full configuration for a specific router, accompanied by explanations to clarify the purpose and function of key commands and sections.](#) Please note that the configurations have been selectively truncated to focus on the most relevant parts for each step.

Following is the initial OSPF configuration in the router Pauillac.

```
router ospf 1
router-id 1.1.1.1
network 10.1.1.0 0.0.0.3 area 0
network 10.1.1.4 0.0.0.3 area 0
network 10.1.1.12 0.0.0.3 area 0
network 10.1.1.16 0.0.0.3 area 0
network 10.10.10.1 0.0.0.0 area 0
```

PART 1: Cisco Express Forwarding (CEF)

1.1 Verify that the router is running a supported Cisco IOS version

```
Pauillac#show version
Cisco IOS Software, 7200 Software (C7200-ADVIPSERVICESK9-M), Version 15.2(4)S5,
RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2014 by Cisco Systems, Inc.
```

Adv-IP-Services-K9 (ADVIPSERVICESK9) router version is a well-known cisco version that supports CEF and MPLS. Hence, the cisco IOS version we used for all of our routers supported CEF and MPLS.

1.2 Verification after configuring CEF with CEF load balancing

Verified that CEF is enabled and successfully populating the FIB (Forwarding Information Base), including equal-cost load-balanced paths for destination prefixes.

```
Pauillac#show ip cef
Prefix      Next Hop      Interface
0.0.0.0/0   no route
0.0.0.0/8   drop
0.0.0.0/32   receive
10.1.1.0/30  attached      Serial1/3
10.1.1.0/32  receive       Serial1/3
10.1.1.1/32  receive       Serial1/3
10.1.1.3/32  receive       Serial1/3
10.1.1.4/30  attached      Serial1/0
10.1.1.4/32  receive       Serial1/0
10.1.1.5/32  receive       Serial1/0
10.1.1.7/32  receive       Serial1/0
10.1.1.8/30  10.1.1.2      Serial1/3
              10.1.1.6      Serial1/0
10.1.1.12/30 attached      Serial1/2
10.1.1.12/32 receive       Serial1/2
10.1.1.13/32 receive       Serial1/2
10.1.1.15/32 receive       Serial1/2
10.1.1.16/30 attached      Serial1/1
10.1.1.16/32 receive       Serial1/1
10.1.1.17/32 receive       Serial1/1
10.1.1.19/32 receive       Serial1/1
Prefix      Next Hop      Interface
10.1.1.20/30 10.1.1.6      Serial1/0
10.10.10.1/32 receive       Loopback0
10.10.10.2/32 10.1.1.2      Serial1/3
10.10.10.3/32 10.1.1.6      Serial1/0
10.10.10.4/32 10.1.1.14     Serial1/2
10.10.10.5/32 10.1.1.18     Serial1/1
10.10.10.6/32 10.1.1.6      Serial1/0
127.0.0.0/8   drop
224.0.0.0/4   drop
224.0.0.0/24  receive
240.0.0.0/4   drop
255.255.255.255/32 receive
```

Confirmed CEF is operational with load-sharing and MPLS support, maintaining active IPv4 and IPv6 tables and optimizing platform-specific forwarding.

```
Pauillac#show cef table
Global information:
  Output chain build favors:
    platform:      not configured
```

```

CLI:          not configured
operational:  convergence-speed
Output chain build characteristics:
Inplace modify
  operational for:  load-sharing
                  push-counter

Collapse
  operational for:  load-sharing

Indirection
  operational for:  recursive-prefix
                  IPv4-to-MPLS
                  IPv6-to-MPLS
                  MPLS-end-of-stack
                  MPLS-non-end-of-stack

MTRIE information:
TAL: node pools:
  pool[C/8 bits]: 14 allocated (0 failed), 14560 bytes {1 refcount}

1 active IPv4 table (32 prefixes total) out of a maximum of 4095.
Table          Prefixes      Memory  Flags
v4:Default          32        18116

1 active IPv6 table (1 prefix total) out of a maximum of 4095.
Table          Prefixes      Memory  Flags
v6:Default          1         208

```

Verified CEF path selection by checking exact forwarding decisions for given source-destination pairs, demonstrating proper interface-based forwarding.

```

Pauillac#show ip cef exact-route 10.1.1.14 10.1.1.6
10.1.1.14 -> 10.1.1.6 => IP adj out of Serial1/0
Pauillac#show ip cef exact-route 10.1.1.18 10.1.1.2
10.1.1.18 -> 10.1.1.2 => IP adj out of Serial1/3
Pauillac#show ip cef exact-route 10.1.1.10 10.1.1.22
10.1.1.10 -> 10.1.1.22 => IP adj out of Serial1/0

```

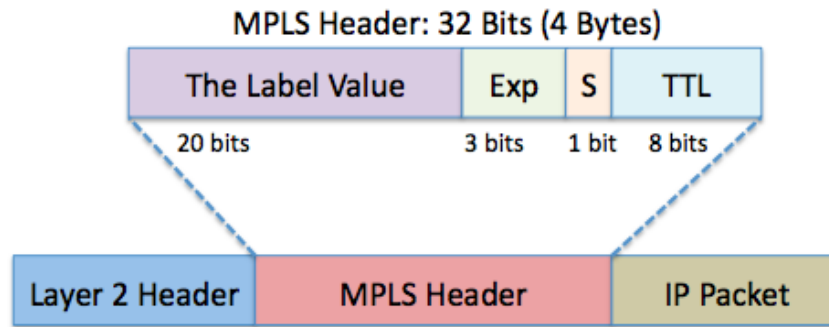
Hence, we can conclude that CEF has been successfully configured.

PART 2: The MPLS Label

2.1 Understanding MPLS

MPLS (Multiprotocol Label Switching) uses short, fixed-length labels to make fast forwarding decisions. Each MPLS label is 32 bits long and contains four fields:

- **Label (20 bits):** Identifies the forwarding equivalence class (FEC).
- **Exp (3 bits):** Used for QoS (e.g., marking traffic priority).
- **S (1 bit):** Bottom-of-stack indicator (used in label stacks).
- **TTL (8 bits):** Time-To-Live to prevent routing loops.



Labels are assigned by **Label Distribution Protocol (LDP)** or other control protocols and are used by MPLS routers (Label Switch Routers) to **switch packets based on the label rather than the IP header**, enabling faster and more efficient routing.

2.2: Assign MPLS Labels

Following is the MPLS configuration in Pauillac

```
Pauillac#show run | section Serial/mpls
interface Serial1/0
 ip address 10.1.1.5 255.255.255.252
 mpls ip
 serial restart-delay 0
interface Serial1/1
 ip address 10.1.1.17 255.255.255.252
 mpls ip
 serial restart-delay 0
interface Serial1/2
 ip address 10.1.1.13 255.255.255.252
 mpls ip
 serial restart-delay 0
interface Serial1/3
 ip address 10.1.1.1 255.255.255.252
 mpls ip
 serial restart-delay 0
 mpls ldp router-id Loopback0 force
```

Verification of MPLS forwarding table

```
Pauillac#show mpls forwarding-table
Local   Outgoing  Prefix      Bytes Label  Outgoing  Next Hop
Label   Label     or Tunnel Id Switched      interface
16      16        10.10.10.6/32 0           Se1/0       point2point
17      Pop Label 10.10.10.4/32 0           Se1/2       point2point
18      Pop Label 10.10.10.3/32 0           Se1/0       point2point
19      Pop Label 10.1.1.20/30  0           Se1/0       point2point
20      Pop Label 10.1.1.8/30   0           Se1/3       point2point
        Pop Label 10.1.1.8/30   0           Se1/0       point2point
21      Pop Label 10.10.10.2/32 0           Se1/3       point2point
22      Pop Label 10.10.10.5/32 0           Se1/1       point2point
```

The above command confirms that MPLS labels are correctly assigned and used to forward packets across interfaces, with some routes using label swapping and others using label popping for penultimate hop popping (PHP).

PART 3: The MPLS Components

3.1 Verification of Label distribution protocol (LDP) bindings

The following command verifies the exchange of MPLS label mappings between routers, showing both local and remote bindings for each prefix, ensuring consistent label allocation across the MPLS network.

```
Pauillac#show mpls ldp bindings
lib entry: 10.1.1.0/30, rev 8
  local binding: label: imp-null
  remote binding: lsr: 10.10.10.4:0, label: 22
  remote binding: lsr: 10.10.10.3:0, label: 20
  remote binding: lsr: 10.10.10.2:0, label: imp-null
  remote binding: lsr: 10.10.10.5:0, label: 23
lib entry: 10.1.1.4/30, rev 2
  local binding: label: imp-null
  remote binding: lsr: 10.10.10.4:0, label: 21
  remote binding: lsr: 10.10.10.3:0, label: imp-null
  remote binding: lsr: 10.10.10.2:0, label: 22
  remote binding: lsr: 10.10.10.5:0, label: 22
lib entry: 10.1.1.8/30, rev 18
  local binding: label: 20
  remote binding: lsr: 10.10.10.4:0, label: 20
  remote binding: lsr: 10.10.10.3:0, label: imp-null
  remote binding: lsr: 10.10.10.2:0, label: imp-null
  remote binding: lsr: 10.10.10.5:0, label: 21
lib entry: 10.1.1.12/30, rev 6
  local binding: label: imp-null
  remote binding: lsr: 10.10.10.4:0, label: imp-null
  remote binding: lsr: 10.10.10.3:0, label: 19
  remote binding: lsr: 10.10.10.2:0, label: 21
  remote binding: lsr: 10.10.10.5:0, label: 20
lib entry: 10.1.1.16/30, rev 4
  local binding: label: imp-null
  remote binding: lsr: 10.10.10.4:0, label: 19
  remote binding: lsr: 10.10.10.3:0, label: 18
  remote binding: lsr: 10.10.10.2:0, label: 20
  remote binding: lsr: 10.10.10.5:0, label: imp-null
lib entry: 10.1.1.20/30, rev 16
  local binding: label: 19
  remote binding: lsr: 10.10.10.4:0, label: 18
  remote binding: lsr: 10.10.10.3:0, label: imp-null
  remote binding: lsr: 10.10.10.2:0, label: 19
  remote binding: lsr: 10.10.10.5:0, label: 19
lib entry: 10.10.10.1/32, rev 20
  local binding: label: imp-null
  remote binding: lsr: 10.10.10.4:0, label: 23
```

```

remote binding: lsr: 10.10.10.3:0, label: 21
remote binding: lsr: 10.10.10.2:0, label: 23
remote binding: lsr: 10.10.10.5:0, label: 24
lib entry: 10.10.10.2/32, rev 22
local binding: label: 21
remote binding: lsr: 10.10.10.2:0, label: imp-null
remote binding: lsr: 10.10.10.3:0, label: 22
remote binding: lsr: 10.10.10.4:0, label: 24
remote binding: lsr: 10.10.10.5:0, label: 25
lib entry: 10.10.10.3/32, rev 14
local binding: label: 18
remote binding: lsr: 10.10.10.4:0, label: 17
remote binding: lsr: 10.10.10.3:0, label: imp-null
remote binding: lsr: 10.10.10.2:0, label: 18
remote binding: lsr: 10.10.10.5:0, label: 18
lib entry: 10.10.10.4/32, rev 12
local binding: label: 17
remote binding: lsr: 10.10.10.4:0, label: imp-null
remote binding: lsr: 10.10.10.3:0, label: 17
remote binding: lsr: 10.10.10.2:0, label: 17
remote binding: lsr: 10.10.10.5:0, label: 17
lib entry: 10.10.10.5/32, rev 24
local binding: label: 22
remote binding: lsr: 10.10.10.4:0, label: 25
remote binding: lsr: 10.10.10.2:0, label: 24
remote binding: lsr: 10.10.10.3:0, label: 23
remote binding: lsr: 10.10.10.5:0, label: imp-null
lib entry: 10.10.10.6/32, rev 10
local binding: label: 16
remote binding: lsr: 10.10.10.4:0, label: 16
remote binding: lsr: 10.10.10.3:0, label: 16
remote binding: lsr: 10.10.10.2:0, label: 16
remote binding: lsr: 10.10.10.5:0, label: 16

```

The following command confirms that LDP sessions with four peer routers are in the operational state, with successful message exchanges over established TCP connections. Each neighbor is discovered via its respective interface and has correctly bound IP addresses for label distribution.

```

Pauillac#show mpls ldp neighbor
Peer LDP Ident: 10.10.10.4:0; Local LDP Ident 10.10.10.1:0
TCP connection: 10.10.10.4.34509 - 10.10.10.1.646
State: Oper; Msgs sent/rcvd: 69/70; Downstream
Up time: 00:47:40
LDP discovery sources:
  Serial1/2, Src IP addr: 10.1.1.14
Addresses bound to peer LDP Ident:
  10.1.1.14      10.10.10.4
Peer LDP Ident: 10.10.10.3:0; Local LDP Ident 10.10.10.1:0
TCP connection: 10.10.10.3.36219 - 10.10.10.1.646
State: Oper; Msgs sent/rcvd: 69/69; Downstream
Up time: 00:47:40
LDP discovery sources:
  Serial1/0, Src IP addr: 10.1.1.6

```

```

Addresses bound to peer LDP Ident:
 10.1.1.6      10.10.10.3      10.1.1.9      10.1.1.21
Peer LDP Ident: 10.10.10.2:0; Local LDP Ident 10.10.10.1:0
TCP connection: 10.10.10.2.18604 - 10.10.10.1.646
State: Oper; Msgs sent/rcvd: 56/56; Downstream
Up time: 00:36:44
LDP discovery sources:
  Serial1/3, Src IP addr: 10.1.1.2
Addresses bound to peer LDP Ident:
 10.1.1.2      10.1.1.10      10.10.10.2
Peer LDP Ident: 10.10.10.5:0; Local LDP Ident 10.10.10.1:0
TCP connection: 10.10.10.5.33638 - 10.10.10.1.646
State: Oper; Msgs sent/rcvd: 51/51; Downstream
Up time: 00:31:32
LDP discovery sources:
  Serial1/1, Src IP addr: 10.1.1.18
Addresses bound to peer LDP Ident:
 10.1.1.18      10.10.10.5

```

3.2 Understanding LER and LSR

- LER (Label Edge Router): A router acts as an LER when it is at the edge of the MPLS domain, performing label imposition (adding a label when sending into MPLS) or label disposition (removing the label when exiting MPLS).
- LSR (Label Switch Router): A router which operates inside the MPLS core, forwarding labeled packets by swapping labels.

From comparing and analyzing MPLS forwarding tables of all 6 ISP routers ([forwarding tables attached in Appendix 9](#)), we can conclude that,

Routers like **Pesaro, Perugia and Pescara** apply(push) outgoing labels such as 16 , 17 ,18, etc., indicating they act as ingress LERs, where labels are first applied to IP packets entering the MPLS domain.

In contrast, routers like **Pescara, Perugia, Pulligny** frequently perform label swapping, (e.g., 17 → 18, 18 → 19) rather than popping which is a key indicator of LSR behavior.

PART 4: The MPLS Process

4.1 Understanding the MPLS Process

MPLS enables efficient and scalable packet forwarding by attaching short labels to packets, which routers use to make forwarding decisions without deep IP header inspection. Labels are distributed between routers using the Label Distribution Protocol (LDP), which ensures that all routers have a consistent view of label-to-prefix mappings. As packets enter the MPLS domain, Label Edge Routers (LERs) assign labels based on destination prefixes. Inside the MPLS core, Label Switch Routers (LSRs) forward packets by swapping labels according to their forwarding tables, and at the egress LER, the label is removed before delivering the packet to its final IP destination.

4.2 Verify MPLS Operation

Following snapshots show the verification of MPLS configs on all routers

```
Pauillac#show mpls interfaces
Interface      IP      Tunnel  BGP  Static  Operational
Serial1/0      Yes (ldp)  No      No   No      Yes
Serial1/1      Yes (ldp)  No      No   No      Yes
Serial1/2      Yes (ldp)  No      No   No      Yes
Serial1/3      Yes (ldp)  No      No   No      Yes

Pomerol#show mpls interfaces
Interface      IP      Tunnel  BGP  Static  Operational
Serial1/0      Yes (ldp)  No      No   No      Yes
Serial1/1      Yes (ldp)  No      No   No      Yes
Serial1/2      Yes (ldp)  No      No   No      Yes

Pulligny#show mpls interfaces
Interface      IP      Tunnel  BGP  Static  Operational
Serial1/0      Yes (ldp)  No      No   No      Yes
Serial1/1      Yes (ldp)  No      No   No      Yes

Perugia#show mpls interfaces
Interface      IP      Tunnel  BGP  Static  Operational
Serial1/0      Yes (ldp)  No      No   No      Yes

Pescara#show mpls interfaces
Interface      IP      Tunnel  BGP  Static  Operational
Serial1/0      Yes (ldp)  No      No   No      Yes

Pesaro#show mpls interfaces
Interface      IP      Tunnel  BGP  Static  Operational
Serial1/0      Yes (ldp)  No      No   No      Yes
```

The following traceroutes confirm end-to-end **MPLS forwarding with proper label assignment**, label swapping at each hop, and label removal at the egress router. They also demonstrate that the network has a functioning **LDP-based label distribution** system, and that MPLS paths are correctly traversed as intended.

```
Perugia#traceroute 10.10.10.6
Type escape sequence to abort.
Tracing the route to 10.10.10.6
VRF info: (vrf in name/id, vrf out name/id)
 1 10.1.1.17 [MPLS: Label 16 Exp 0] 60 msec 28 msec 28 msec
 2 10.1.1.6 [MPLS: Label 16 Exp 0] 32 msec 20 msec 20 msec
 3 10.1.1.22 40 msec 32 msec 28 msec
Perugia#traceroute 10.1.1.22
Type escape sequence to abort.
Tracing the route to 10.1.1.22
VRF info: (vrf in name/id, vrf out name/id)
 1 10.1.1.17 [MPLS: Label 19 Exp 0] 12 msec 8 msec 12 msec
 2 10.1.1.6 20 msec 20 msec 20 msec
 3 10.1.1.22 28 msec 32 msec 28 msec
```

```

Pescara#traceroute 10.10.10.6
Type escape sequence to abort.
Tracing the route to 10.10.10.6
VRF info: (vrf in name/id, vrf out name/id)
 1 10.1.1.13 [MPLS: Label 16 Exp 0] 28 msec 28 msec 32 msec
 2 10.1.1.6 [MPLS: Label 16 Exp 0] 20 msec 20 msec 20 msec
 3 10.1.1.22 28 msec 32 msec 28 msec
Pescara#traceroute 10.1.1.22
Type escape sequence to abort.
Tracing the route to 10.1.1.22
VRF info: (vrf in name/id, vrf out name/id)
 1 10.1.1.13 [MPLS: Label 19 Exp 0] 4 msec 8 msec 8 msec
 2 10.1.1.6 20 msec 20 msec 20 msec
 3 10.1.1.22 32 msec 28 msec 32 msec

Pesaro#traceroute 10.10.10.4
Type escape sequence to abort.
Tracing the route to 10.10.10.4
VRF info: (vrf in name/id, vrf out name/id)
 1 10.1.1.21 [MPLS: Label 17 Exp 0] 40 msec 36 msec 40 msec
 2 10.1.1.5 [MPLS: Label 17 Exp 0] 32 msec 28 msec 28 msec
 3 10.1.1.14 40 msec 40 msec 40 msec
Pesaro#traceroute 10.10.10.5
Type escape sequence to abort.
Tracing the route to 10.10.10.5
VRF info: (vrf in name/id, vrf out name/id)
 1 10.1.1.21 [MPLS: Label 23 Exp 0] 24 msec 32 msec 28 msec
 2 10.1.1.5 [MPLS: Label 22 Exp 0] 20 msec 20 msec 20 msec
 3 10.1.1.18 32 msec 32 msec 28 msec
Pesaro#traceroute 10.1.1.14
Type escape sequence to abort.
Tracing the route to 10.1.1.14
VRF info: (vrf in name/id, vrf out name/id)
 1 10.1.1.21 [MPLS: Label 19 Exp 0] 16 msec 16 msec 20 msec
 2 10.1.1.5 28 msec 32 msec 28 msec
 3 10.1.1.14 40 msec 40 msec 40 msec
Pesaro#traceroute 10.1.1.18
Type escape sequence to abort.
Tracing the route to 10.1.1.18
VRF info: (vrf in name/id, vrf out name/id)
 1 10.1.1.21 [MPLS: Label 18 Exp 0] 4 msec 12 msec 8 msec
 2 10.1.1.5 20 msec 20 msec 20 msec
 3 10.1.1.18 32 msec 28 msec 32 msec

```

PART 5: Configuring Basic MPLS Traffic Engineering

A tunnel was created between Perugia and Pesaro Provider edges.

Following verifies that an MPLS TE tunnel is up and actively forwarding traffic.

```

Paulliac#show mpls traffic-eng tunnels

P2P TUNNELS/LSPs:

```

```

LSP Tunnel Perugia_t0 is signalled, connection is up
InLabel  : Serial1/1, 23
Prev Hop : 10.1.1.18
OutLabel  : Serial1/0, 24
Next Hop : 10.1.1.6
RSVP Signalling Info:
  Src 10.10.10.5, Dst 10.10.10.6, Tun_Id 0, Tun_Instance 1
RSVP Path Info:
  My Address: 10.1.1.5
  Explicit Route: 10.1.1.6 10.1.1.22 10.10.10.6
  Record Route: NONE
  Tspec: ave rate=0 kbits, burst=1000 bytes, peak rate=0 kbits
RSVP Resv Info:
  Record Route: NONE
  Tspec: ave rate=0 kbits, burst=1000 bytes, peak rate=0 kbits

P2MP TUNNELS:

P2MP SUB-LSPS:

```

In the routing table as shown below, the Tunnel appears as a separate routing entry.

```

Perugia#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

10.0.0.0/8 is variably subnetted, 13 subnets, 2 masks
O    10.1.1.0/30 [110/128] via 10.1.1.17, 03:16:30, Serial1/0
O    10.1.1.4/30 [110/128] via 10.1.1.17, 03:16:30, Serial1/0
O    10.1.1.8/30 [110/192] via 10.1.1.17, 03:16:30, Serial1/0
O    10.1.1.12/30 [110/128] via 10.1.1.17, 03:16:30, Serial1/0
C    10.1.1.16/30 is directly connected, Serial1/0
L    10.1.1.18/32 is directly connected, Serial1/0
O    10.1.1.20/30 [110/192] via 10.1.1.17, 03:16:30, Serial1/0
O    10.10.10.1/32 [110/65] via 10.1.1.17, 02:07:12, Serial1/0
O    10.10.10.2/32 [110/129] via 10.1.1.17, 01:57:39, Serial1/0
O    10.10.10.3/32 [110/129] via 10.1.1.17, 03:16:30, Serial1/0
O    10.10.10.4/32 [110/129] via 10.1.1.17, 03:16:30, Serial1/0
C    10.10.10.5/32 is directly connected, Loopback0
O    10.10.10.6/32 [110/193] via 10.10.10.6, 00:02:15, Tunnel0

```

As shown in the following, CEF chooses best path as the tunnel for the destination

```

Perugia#show ip cef 10.10.10.6
10.10.10.6/32
  nexthop 10.10.10.6 Tunnel0

```

The tunnel can be observed at the MPLS forwarding table as well, as shown below.

```
Perugia#show mpls forwarding-table
Local   Outgoing   Prefix      Bytes Label  Outgoing   Next Hop
Label   Label      or Tunnel Id Switched      interface
16      [T] Pop Label 10.10.10.6/32 0             Tu0        point2point
17      17         10.10.10.4/32 0             Se1/0      point2point
18      18         10.10.10.3/32 0             Se1/0      point2point
19      19         10.1.1.20/30  0             Se1/0      point2point
20      Pop Label 10.1.1.12/30  0             Se1/0      point2point
21      20         10.1.1.8/30   0             Se1/0      point2point
22      Pop Label 10.1.1.4/30   0             Se1/0      point2point
23      Pop Label 10.1.1.0/30   0             Se1/0      point2point
24      Pop Label 10.10.10.1/32 0             Se1/0      point2point
25      21         10.10.10.2/32 0             Se1/0      point2point

[T] Forwarding through a LSP tunnel.
View additional labelling info with the 'detail' option
```

The following traceroute shows the actual forwarding path and verifies label-based routing is working.

- Packets follow the MPLS tunnel with labels **23** → **24**, as defined by the TE tunnel.
- The labels align with the **RSVP-signaled LSP path** seen earlier, proving that **TE constraints are respected** in live traffic.

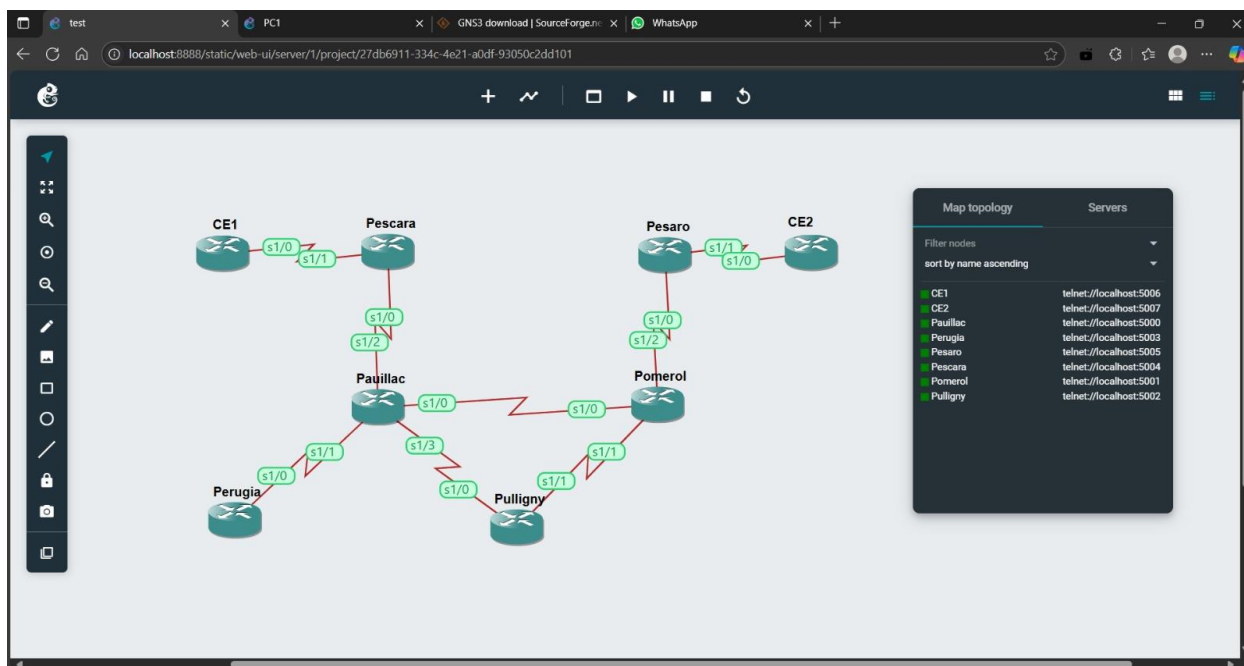
```
Perugia#traceroute 10.10.10.6
Type escape sequence to abort.
Tracing the route to 10.10.10.6
VRF info: (vrf in name/id, vrf out name/id)
 1 10.1.1.17 [MPLS: Label 23 Exp 0] 32 msec 40 msec 40 msec
 2 10.1.1.6 [MPLS: Label 24 Exp 0] 28 msec 32 msec 28 msec
 3 10.1.1.22 32 msec 28 msec 32 msec
```

PART 6: MPLS VPNs

6.1 Configure MPLS VPN

VPN tunnel was created between Pescara and Pesaro.

Additional two routers (Customer Edge Routers: CE1, CE2) were added as following for ease of demonstrating a VPN network. They were connected to the provide edge routers Pescara and Pesaro respectively.



Verification of the VPN through routing table

```
Pesaro#show ip route vrf CUSTOMER1

Routing Table: CUSTOMER1
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

B    192.168.1.0/24 [200/0] via 10.10.10.4, 00:29:22
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.2.0/24 is directly connected, Serial1/1
L    192.168.2.1/32 is directly connected, Serial1/1
```

As shown above;

Route to remote Customer site 1 is learned via BGP over the MPLS backbone.
(Pesaro is receiving this VPNv4 route from a PE connected to 10.10.10.4.)

Route to local customer subnet connected directly on interface Serial1/1.

This confirms Pesaro is a PE router for CUSTOMER1, with a directly connected CE (Customer Edge) device.

The below snapshot shows, CEF forwarding entries for CUSTOMER1's VRF.

- 192.168.1.0/24 → 10.1.1.21 Serial1/0:
Traffic to remote site (192.168.1.0/24) goes out via MPLS core over Serial1/0.
- 192.168.2.0/24 → attached Serial1/1:
Local customer subnet directly connected.
- This verifies MPLS forwarding is set up properly, with remote VPN prefixes reachable.

```
Pesaro#show ip cef vrf CUSTOMER1
Prefix          Next Hop          Interface
0.0.0.0/0       no route
0.0.0.0/8       drop
0.0.0.0/32      receive
127.0.0.0/8     drop
192.168.1.0/24  10.1.1.21        Serial1/0
192.168.2.0/24  attached         Serial1/1
192.168.2.0/32  receive         Serial1/1
192.168.2.1/32  receive         Serial1/1
192.168.2.255/32 receive         Serial1/1
224.0.0.0/4     drop
224.0.0.0/24    receive
240.0.0.0/4     drop
255.255.255.255/32 receive
```

Traceroute from **Customer Edge 2 (CE2)** to **Customer Edge 1 (CE1)** (192.168.1.2) was carried out as follows to test the VPN connection.

```
CE2#traceroute 192.168.1.2
Type escape sequence to abort.
Tracing the route to 192.168.1.2
VRF info: (vrf in name/id, vrf out name/id)
 1 192.168.2.1 8 msec 12 msec 8 msec
 2 10.1.1.21 [MPLS: Labels 18/26 Exp 0] 40 msec 40 msec 52 msec
 3 10.1.1.5 [MPLS: Labels 18/26 Exp 0] 52 msec 40 msec 40 msec
 4 192.168.1.1 48 msec 40 msec 40 msec
 5 192.168.1.2 60 msec 52 msec 52 msec
```

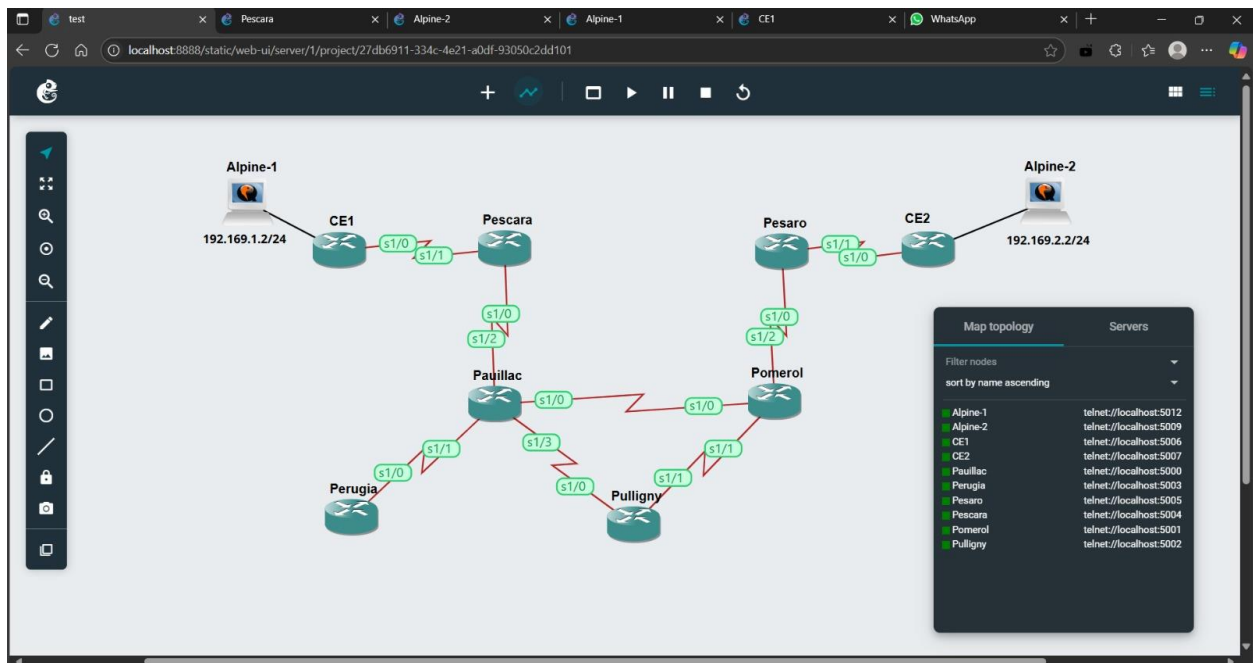
- Hop 1: 192.168.2.1 → local PE router (Pesaro).
- Hop 2 & 3: MPLS-labeled hops over core (labels 18/26) – showing VPN label stacking:
 - Inner label (26): identifies the VPN route (192.168.1.0).
 - Outer label (18): used for LSP (Label Switched Path) across the backbone.
- Hop 4 & 5: CE1 router and host.

This proves that end-to-end MPLS VPN forwarding is working, carrying traffic across PE routers using BGP VPNv4 and label stacking.

The final and complete routing configs of CE1 and CE2 are attached in Appendices 7,8.

6.2 Configure QoS

To generate VOIP and Video traffic, customer hosts (PC1 and PC2 which are Alpine linux VMs with iperf3 installed) were connected to customer edge routers CE1 and CE2 respectively as follows.



Alpine-1 was setup as iperf3 client while Alpine-2 was the iperf3 server for VOIP and Video. Since, generally the first ISP router applies QoS, Pescara's ingress port was used to enforce QoS policies.

A QoS policy named MPLS-QOS was configured and applied on the input of interface Serial1/1 of Pescara, which connects the PE router(Pesara) to the CE. The policy classifies traffic into three classes: VOICE-IN (matching DSCP EF for voice), VIDEO-IN (matching DSCP AF41 for video), and a default class for all other traffic. The statistics show that voice and video traffic were correctly matched and conformed to their configured CIR (1 Mbps for voice, 2 Mbps for video)

with no packet drops, confirming that QoS is functioning as expected to prioritize critical traffic in the MPLS VPN. Following screenshot shows QoS statistics at ingress port of PE router Pescara.

```
Pescara#show policy-map interface serial 1/1
Serial1/1

Service-policy input: MPLS-QOS

Class-map: VOICE-IN (match-any)
  4526 packets, 868680 bytes
  5 minute offered rate 0000 bps, drop rate 0000 bps
  Match: dscp ef (46)
    4526 packets, 868680 bytes
    5 minute rate 0 bps
  police:
    cir 1000000 bps, bc 8000 bytes
    conformed 4526 packets, 868680 bytes; actions:
      transmit
    exceeded 0 packets, 0 bytes; actions:
      drop
    conformed 0000 bps, exceeded 0000 bps

Class-map: VIDEO-IN (match-any)
  773 packets, 523333 bytes
  5 minute offered rate 15000 bps, drop rate 0000 bps
  Match: dscp af41 (34)
    773 packets, 523333 bytes
    5 minute rate 15000 bps
  police:
    cir 2000000 bps, bc 8000 bytes
    conformed 773 packets, 523333 bytes; actions:
      transmit
    exceeded 0 packets, 0 bytes; actions:
      drop
    conformed 15000 bps, exceeded 0000 bps

Class-map: class-default (match-any)
  138 packets, 11541 bytes
  5 minute offered rate 0000 bps, drop rate 0000 bps
  Match: any
  QoS Set
    dscp default
    Packets marked 138
```

Hence, it can be concluded that the configured QoS policies on VPN guarantees bandwidth reservation and prioritization, meeting QoS goals.

The above tests used iperf3 with --tos 184 for voice (EF) and --tos 136 for video (AF41).

The VOIP server-client connection (EF == DSCP 46 == --tos 184)

```
alpine1:~# iperf3 -c 192.169.2.2 -u -b 64K -l 160 -t 60 --tos 184
Connecting to host 192.169.2.2, port 5201
[ 5] local 192.169.1.2 port 60696 connected to 192.169.2.2 port 5201
[ ID] Interval            Transfer        Bitrate          Total Datagrams
[ 5] 0.00-1.00 sec      7.81 KBytes    64.0 Kbits/sec   50
[ 5] 1.00-2.00 sec      7.81 KBytes    64.0 Kbits/sec   50
[ 5] 2.00-3.00 sec      7.81 KBytes    64.0 Kbits/sec   50
[ 5] 3.00-4.00 sec      7.81 KBytes    64.0 Kbits/sec   50
[ 5] 4.00-5.00 sec      7.81 KBytes    64.0 Kbits/sec   50
[ 5] 5.00-6.00 sec      7.81 KBytes    64.0 Kbits/sec   50
[ 5] 6.00-7.00 sec      7.81 KBytes    64.0 Kbits/sec   50
[ 5] 7.00-8.00 sec      7.81 KBytes    64.0 Kbits/sec   50
```

```
Server listening on 5201 (test #2)
-----
Accepted connection from 192.169.1.2, port 42436
[ 5] local 192.169.2.2 port 5201 connected to 192.169.1.2 port 60696
[ ID] Interval            Transfer        Bitrate          Jitter    Lost/Total Datagrams
[ 5] 0.00-1.00 sec      7.34 KBytes    60.1 Kbits/sec   0.170 ms  0/47 (0%)
[ 5] 1.00-2.00 sec      7.81 KBytes    64.0 Kbits/sec   0.945 ms  0/50 (0%)
[ 5] 2.00-3.00 sec      7.81 KBytes    64.0 Kbits/sec   0.281 ms  0/50 (0%)
[ 5] 3.00-4.00 sec      7.81 KBytes    64.0 Kbits/sec   0.234 ms  0/50 (0%)
[ 5] 4.00-5.00 sec      7.81 KBytes    64.0 Kbits/sec   0.812 ms  0/50 (0%)
```

The Video server-client connection (AF41 == DSCP 34 == --tos 136)

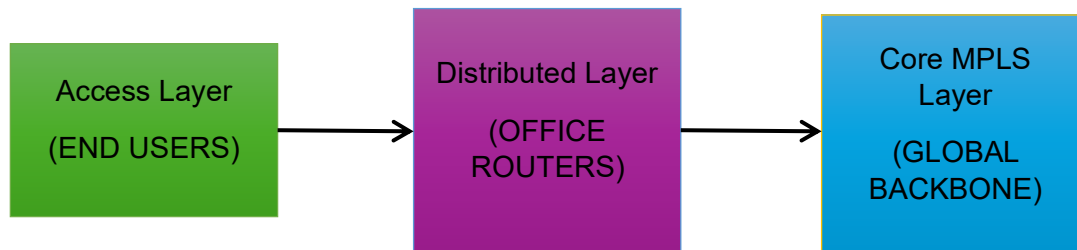
```
alpine1:~# iperf3 -c 192.169.2.2 -b 64K -l 160 -t 60 --tos 136
Connecting to host 192.169.2.2, port 5201
[ 5] local 192.169.1.2 port 48968 connected to 192.169.2.2 port 5201
[ ID] Interval            Transfer        Bitrate          Retr    Cwnd
[ 5] 0.00-1.00 sec      7.81 KBytes    64.0 Kbits/sec    0    14.1 KBytes
[ 5] 1.00-2.00 sec      7.81 KBytes    64.0 Kbits/sec    0    14.1 KBytes
[ 5] 2.00-3.00 sec      7.81 KBytes    64.0 Kbits/sec    0    14.1 KBytes
[ 5] 3.00-4.00 sec      7.81 KBytes    64.0 Kbits/sec    0    14.1 KBytes
[ 5] 4.00-5.00 sec      7.81 KBytes    64.0 Kbits/sec    0    14.1 KBytes
```

```
-----
Server listening on 5201 (test #4)
-----
Accepted connection from 192.169.1.2, port 48966
[ 5] local 192.169.2.2 port 5201 connected to 192.169.1.2 port 48968
[ ID] Interval            Transfer        Bitrate
[ 5] 0.00-1.00 sec      7.34 KBytes    60.1 Kbits/sec
[ 5] 1.00-2.00 sec      7.66 KBytes    62.7 Kbits/sec
[ 5] 2.00-3.00 sec      7.66 KBytes    62.7 Kbits/sec
[ 5] 3.00-4.00 sec      7.81 KBytes    64.0 Kbits/sec
[ 5] 4.00-5.00 sec      7.97 KBytes    65.3 Kbits/sec
```

Case Study: XYZ Corp MPLS Network Architecture

1. High-Level Overview

The network is built in three hierarchical layers:



2. Core Layer - MPLS Backbone

Purpose: High-speed, scalable inter-office connectivity

- Location: Assuming globally interconnects USA, Germany, and Japan offices
- Routers: Label Switch Routers (LSRs) - only switch MPLS-labeled packets
- Protocols:
 - IGP: OSPF or IS-IS
 - LDP: For label distribution of MPLS
 - RSVP-TE: For traffic engineering
- Technologies:
 - Cisco Express Forwarding (CEF)
 - MPLS VPNs for secure inter-branch communication
 - QoS policies to prioritize real-time traffic (voice/video)

3. Distribution Layer – Office-Level Routing

Purpose: Connect each local office to the core MPLS cloud

- Each country (USA, Germany, Japan) has:
 - A Distribution Router or Label Edge Router (LER)
 - Performs label imposition (ingress) and disposition (egress)
 - VRFs (Virtual Routing and Forwarding) configured per department

- Functions:
 - Enforces MPLS VPN policies
 - Performs routing between departments and the core
 - Applies QoS classification and marking

4. Access Layer – Department-Level Network

Purpose: Connect internal devices and departments to the network

- Departments:
 - Accounting
 - Operations
 - HR & Administration
 - Sales
- Each department:
 - Resides in its own VLAN
 - Connected to the access switches
 - Assigned to a specific MPLS VPN for logical separation
- Access Layer connects:
 - End devices (PCs, phones, printers)
 - IP phones and video devices (QoS critical)

5. Centralized Data Center

Location: USA (can be redundant in another country later)

Connected via:

- High-speed MPLS TE tunnel with QoS priority
- Accessible securely by all departments in all countries
- Hosts:
 - File servers
 - CRM, ERP
 - Email servers
 - Cloud gateway

6. Security & Performance Features

<i>Feature</i>	<i>Description</i>
<i>MPLS VPNs</i>	Separate logical networks per department/site
<i>QoS (Quality of Service)</i>	Prioritize VoIP and video conferencing using DSCP/EXP bits
<i>CEF (Cisco Express Forwarding)</i>	Fast packet lookup and forwarding
<i>Traffic Engineering (RSVP-TE)</i>	Optimizes path selection and load balancing across MPLS cloud
<i>Traceroute / Monitoring</i>	Tracks MPLS paths and performance metrics

Appendix 1: Core Router – Pauillac

This ISP core router Pauillac is configured as a MPLS-enabled backbone router participating in OSPF for routing and LDP for label distribution. MPLS Traffic Engineering (TE) is enabled on all active serial interfaces with RSVP-based bandwidth reservation. The router uses Loopback0 (10.10.10.1) as its router ID for both OSPF and MPLS LDP, and CEF is enabled to support high-speed packet forwarding.

```
Pauillac#show run
Building configuration...

Current configuration : 2533 bytes
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
!
hostname Pauillac
!
boot-start-marker
boot-end-marker
!
no aaa new-model
no ip icmp rate-limit unreachable
ip cef
!
no ip domain lookup
no ipv6 cef
!
!
mpls label protocol ldp
mpls traffic-eng tunnels
multilink bundle-name authenticated
!
ip tcp synwait-time 5
!
!
interface Loopback0
 ip address 10.10.10.1 255.255.255.255
!
interface FastEthernet0/0
 no ip address
 shutdown
 duplex full
!
interface Serial1/0
 ip address 10.1.1.5 255.255.255.252
 mpls ip
 mpls traffic-eng tunnels
 serial restart-delay 0
 ip rsvp bandwidth 1000000 1000000
!
```

```

interface Serial1/1
 ip address 10.1.1.17 255.255.255.252
 mpls ip
 mpls traffic-eng tunnels
 serial restart-delay 0
 ip rsvp bandwidth 1000000 1000000
!
interface Serial1/2
 ip address 10.1.1.13 255.255.255.252
 mpls ip
 mpls traffic-eng tunnels
 serial restart-delay 0
 ip rsvp bandwidth 1000000 1000000
!
interface Serial1/3
 ip address 10.1.1.1 255.255.255.252
 mpls ip
 mpls traffic-eng tunnels
 serial restart-delay 0
 ip rsvp bandwidth 1000000 1000000
!
!
router ospf 1
 router-id 1.1.1.1
 network 10.1.1.0 0.0.0.3 area 0
 network 10.1.1.4 0.0.0.3 area 0
 network 10.1.1.12 0.0.0.3 area 0
 network 10.1.1.16 0.0.0.3 area 0
 network 10.10.10.1 0.0.0.0 area 0
 mpls traffic-eng router-id Loopback0
 mpls traffic-eng area 0
!
ip forward-protocol nd
!
mpls ldp router-id Loopback0 force
!
!
control-plane
!
!
line con 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
 stopbits 1
line aux 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
 stopbits 1
line vty 0 4
 login
!
!
end

```

Appendix 2: Core Router – Pomerol

This core router Pomerol is configured as an MPLS backbone router using OSPF for IGP and LDP for MPLS label distribution. It has CEF enabled for fast packet switching and supports MPLS Traffic Engineering via RSVP on all active serial links. The router uses Loopback0 (10.10.10.3) as the MPLS and OSPF router ID, and it participates in area 0 for OSPF. This setup allows it to efficiently forward labeled packets and support MPLS VPN and TE paths in the ISP core.

```
Pomerol#show run
Building configuration...

Current configuration : 2415 bytes
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
!
hostname Pomerol
!
boot-start-marker
boot-end-marker
!
no aaa new-model
no ip icmp rate-limit unreachable
ip cef
!
no ip domain lookup
no ipv6 cef
!
!
mpls label protocol ldp
mpls traffic-eng tunnels
multilink bundle-name authenticated
!
ip tcp synwait-time 5
!
!
interface Loopback0
 ip address 10.10.10.3 255.255.255.255
!
interface FastEthernet0/0
 no ip address
 shutdown
 duplex full
!
interface Serial1/0
 ip address 10.1.1.6 255.255.255.252
 mpls ip
 mpls traffic-eng tunnels
 serial restart-delay 0
 ip rsvp bandwidth 1000000 1000000
!
```

```

interface Serial1/1
 ip address 10.1.1.9 255.255.255.252
 mpls ip
 mpls traffic-eng tunnels
 serial restart-delay 0
 ip rsvp bandwidth 1000000 1000000
!
interface Serial1/2
 ip address 10.1.1.21 255.255.255.252
 mpls ip
 mpls traffic-eng tunnels
 serial restart-delay 0
 ip rsvp bandwidth 1000000 1000000
!
interface Serial1/3
 no ip address
 shutdown
 serial restart-delay 0
!
!
router ospf 1
 router-id 3.3.3.3
 network 10.1.1.4 0.0.0.3 area 0
 network 10.1.1.8 0.0.0.3 area 0
 network 10.1.1.20 0.0.0.3 area 0
 network 10.10.10.3 0.0.0.0 area 0
 mpls traffic-eng router-id Loopback0
 mpls traffic-eng area 0
!
ip forward-protocol nd
!
!
mpls ldp router-id Loopback0 force
!
!
control-plane
!
!
line con 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
 stopbits 1
line aux 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
 stopbits 1
line vty 0 4
 login
!
!
end

```


Appendix 3: Core Router – Pulligny

This core router Pulligny is also configured as an MPLS backbone node with CEF enabled for efficient forwarding and OSPF for IGP routing in area 0. It uses Loopback0 (10.10.10.2) as the router ID for both OSPF and MPLS LDP. MPLS is enabled on the active serial links with RSVP-based bandwidth reservation to support Traffic Engineering (TE). This setup allows Pulligny to participate in MPLS label switching and TE path calculations within the ISP core.

```
Pulligny#show run
Building configuration...

Current configuration : 2300 bytes
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
!
!
hostname Pulligny
!
boot-start-marker
boot-end-marker
!
no aaa new-model
no ip icmp rate-limit unreachable
ip cef
!
no ip domain lookup
no ipv6 cef
!
!
mpls label protocol ldp
mpls traffic-eng tunnels
multilink bundle-name authenticated
!
ip tcp synwait-time 5
!
!
interface Loopback0
 ip address 10.10.10.2 255.255.255.255
!
interface FastEthernet0/0
 no ip address
 shutdown
 duplex full
!
interface Serial1/0
 ip address 10.1.1.2 255.255.255.252
 mpls ip
 mpls traffic-eng tunnels
 serial restart-delay 0
 ip rsvp bandwidth 1000000 1000000
!
```

```

interface Serial1/1
 ip address 10.1.1.10 255.255.255.252
 mpls ip
 mpls traffic-eng tunnels
 serial restart-delay 0
 ip rsvp bandwidth 1000000 1000000
!
!
router ospf 1
 router-id 2.2.2.2
 network 10.1.1.0 0.0.0.3 area 0
 network 10.1.1.8 0.0.0.3 area 0
 network 10.10.10.2 0.0.0.0 area 0
 mpls traffic-eng router-id Loopback0
 mpls traffic-eng area 0
!
ip forward-protocol nd
!
!
mpls ldp router-id Loopback0 force
!
!
control-plane
!
!
line con 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
 stopbits 1
line aux 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
 stopbits 1
line vty 0 4
 login
!
!
end

```

Appendix 4: Provider Edge Router – Perugia

This provider edge (PE) router - Perugia is configured to operate at the edge of the MPLS network, participating in OSPF for routing and LDP for MPLS label distribution. It has CEF and MPLS Traffic Engineering enabled, and acts as a tunnel head-end using Tunnel0 to establish a dynamic TE tunnel toward 10.10.10.6. The tunnel is configured to use Loopback0 as its source and advertises routes through the tunnel using autoroute announce. This allows Perugia to direct selected traffic over optimized MPLS TE paths within the core.

```
Perugia#show run
Building configuration...

Current configuration : 2382 bytes
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
!
hostname Perugia
!
boot-start-marker
boot-end-marker
!
no aaa new-model
no ip icmp rate-limit unreachable
ip cef
!
no ip domain lookup
no ipv6 cef
!
!
mpls label protocol ldp
mpls traffic-eng tunnels
multilink bundle-name authenticated
!
ip tcp synwait-time 5
!
!
interface Loopback0
 ip address 10.10.10.5 255.255.255.255
!
interface Tunnel0
 ip unnumbered Loopback0
 tunnel mode mpls traffic-eng
 tunnel destination 10.10.10.6
 tunnel mpls traffic-eng autoroute announce
 tunnel mpls traffic-eng path-option 1 dynamic
!
interface FastEthernet0/0
 no ip address
 shutdown
 duplex full
!
```

```

interface Serial1/0
 ip address 10.1.1.18 255.255.255.252
 mpls ip
 mpls traffic-eng tunnels
 serial restart-delay 0
 ip rsvp bandwidth 1000000 1000000
!
interface Serial1/1
 no ip address
 shutdown
 serial restart-delay 0
!
!
router ospf 1
 router-id 5.5.5.5
 network 10.1.1.16 0.0.0.3 area 0
 network 10.10.10.5 0.0.0.0 area 0
 mpls traffic-eng router-id Loopback0
 mpls traffic-eng area 0
!
ip forward-protocol nd
!
!
mpls ldp router-id Loopback0 force
!
!
control-plane
!
!
line con 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
 stopbits 1
line aux 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
 stopbits 1
line vty 0 4
 login
!
!
end

```

Appendix 5: Provider Edge Router – Pescara

The Pescara router acts as a Provider Edge (PE) router supporting both MPLS VPN and QoS services. It defines a VRF named CUSTOMER1 with route distinguishers and import/export route targets for VPN separation. It connects to the MPLS core via Serial1/0 and to the customer via Serial1/1, which is bound to the VRF. BGP is configured to exchange VPNv4 routes with another PE router over loopback addresses. It also applies a QoS policy (MPLS-QOS) to prioritize voice (EF) and video (AF41) traffic using input policing. This configuration enables secure, isolated VPN routing for the customer while enforcing traffic prioritization on ingress.

```
Pescara#show run
Building configuration...

Current configuration : 3236 bytes
!
! Last configuration change at 13:03:41 UTC Sat Jul 12 2025
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
!
hostname Pescara
!
boot-start-marker
boot-end-marker
!
no aaa new-model
no ip icmp rate-limit unreachable
ip cef
!
!
ip vrf CUSTOMER1
  rd 100:1
  route-target export 100:1
  route-target import 100:1
!
no ip domain lookup
no ipv6 cef
!
!
mpls label protocol ldp
mpls traffic-eng tunnels
multilink bundle-name authenticated
!
!
ip tcp synwait-time 5
```

```

!
class-map match-any VIDEO-IN
  match dscp af41
class-map match-any VOICE-IN
  match dscp ef
!
policy-map MPLS-QOS
  class VOICE-IN
    police 1000000 8000 conform-action transmit exceed-action drop
  class VIDEO-IN
    police 2000000 8000 conform-action transmit exceed-action drop
  class class-default
    set dscp default
!
interface Loopback0
  ip address 10.10.10.4 255.255.255.255
!
interface FastEthernet0/0
  no ip address
  shutdown
  duplex full
!
interface Serial1/0
  ip address 10.1.1.14 255.255.255.252
  mpls ip
  mpls traffic-eng tunnels
  serial restart-delay 0
  ip rsvp bandwidth 1000000 1000000
!
interface Serial1/1
  ip vrf forwarding CUSTOMER1
  ip address 192.168.1.1 255.255.255.0
  serial restart-delay 0
  service-policy input MPLS-QOS
!
router ospf 100 vrf CUSTOMER1
  redistribute bgp 65000 subnets
  network 192.168.1.0 0.0.0.255 area 0
!
router ospf 1
  router-id 4.4.4.4
  network 10.1.1.12 0.0.0.3 area 0
  network 10.10.10.4 0.0.0.0 area 0
  mpls traffic-eng router-id Loopback0
  mpls traffic-eng area 0

```

```
!  
router bgp 65000  
  bgp log-neighbor-changes  
  no bgp default ipv4-unicast  
  neighbor 10.10.10.6 remote-as 65000  
  neighbor 10.10.10.6 update-source Loopback0  
  !  
  address-family ipv4  
  exit-address-family  
  !  
  address-family vpnv4  
    neighbor 10.10.10.6 activate  
    neighbor 10.10.10.6 send-community extended  
  exit-address-family  
  !  
  address-family ipv4 vrf CUSTOMER1  
    redistribute ospf 100  
  exit-address-family  
  !  
ip forward-protocol nd  
!  
!  
mpls ldp router-id Loopback0 force  
!  
!  
control-plane  
!  
!  
line con 0  
  exec-timeout 0 0  
  privilege level 15  
  logging synchronous  
  stopbits 1  
line aux 0  
  exec-timeout 0 0  
  privilege level 15  
  logging synchronous  
  stopbits 1  
line vty 0 4  
  login  
!  
!  
end
```

Appendix 6: Provider Edge Router – Pesaro

The Pesaro router is a Provider Edge (PE) router and serves as the tail end of an MPLS TE tunnel from Perugia. It supports MPLS VPNs via the CUSTOMER1 VRF, with BGP and OSPF configured to exchange customer routes. Interface Serial1/1 is VRF-bound and connects to the CE router, while Serial1/0 links to the MPLS core with MPLS and RSVP-TE enabled. Pesaro participates in the MPLS core using OSPF and LDP, and advertises VPN routes using MP-BGP with VPNv4 address family. This setup enables isolated, end-to-end VPN connectivity with MPLS and traffic engineering support.

```
Pesaro#show run
Building configuration...

Current configuration : 2880 bytes
!
! Last configuration change at 13:30:42 UTC Sat Jul 12 2025
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
!
hostname Pesaro
!
boot-start-marker
boot-end-marker
!
no aaa new-model
no ip icmp rate-limit unreachable
ip cef
!
ip vrf CUSTOMER1
  rd 100:1
  route-target export 100:1
  route-target import 100:1
!
no ip domain lookup
no ipv6 cef
!
!
mpls label protocol ldp
mpls traffic-eng tunnels
multilink bundle-name authenticated
!
ip tcp synwait-time 5
!
!
interface Loopback0
  ip address 10.10.10.6 255.255.255.255
!
interface FastEthernet0/0
  no ip address
  shutdown
  duplex full
```



```

!
interface Serial1/0
 ip address 10.1.1.22 255.255.255.252
 mpls ip
 mpls traffic-eng tunnels
 serial restart-delay 0
 ip rsvp bandwidth 1000000 1000000
!
interface Serial1/1
 ip vrf forwarding CUSTOMER1
 ip address 192.168.2.1 255.255.255.0
 serial restart-delay 0
!
!
router ospf 100 vrf CUSTOMER1
 redistribute bgp 65000 subnets
 network 192.168.2.0 0.0.0.255 area 0
!
router ospf 1
 router-id 6.6.6.6
 network 10.1.1.20 0.0.0.3 area 0
 network 10.10.10.6 0.0.0.0 area 0
 mpls traffic-eng router-id Loopback0
 mpls traffic-eng area 0
!
router bgp 65000
 bgp log-neighbor-changes
 no bgp default ipv4-unicast
 neighbor 10.10.10.4 remote-as 65000
 neighbor 10.10.10.4 update-source Loopback0
!
 address-family ipv4
  exit-address-family
!
 address-family vpnv4
  neighbor 10.10.10.4 activate
  neighbor 10.10.10.4 send-community extended
  exit-address-family
!
 address-family ipv4 vrf CUSTOMER1
  redistribute ospf 100
  exit-address-family
!
ip forward-protocol nd
!
!
!
mpls ldp router-id Loopback0 force
!
!
control-plane
!
!
line con 0
 exec-timeout 0 0

```

```
privilege level 15
logging synchronous
stopbits 1
line aux 0
exec-timeout 0 0
privilege level 15
logging synchronous
stopbits 1
line vty 0 4
login
!
!
end
```

Appendix 7: Customer Edge Router – CE1

The CE1 router acts as the Customer Edge router connecting an enterprise site to the MPLS VPN via the Pescara PE router. It uses OSPF for dynamic routing with the provider (Serial1/0) and to its internal LAN (FastEthernet0/0). The router does not run MPLS or VRF, as typical for CE devices. Its main role is to exchange customer routes with the PE, enabling site-to-site VPN connectivity through the provider's MPLS network. This setup keeps the CE simple, relying on the PE for VPN segmentation and MPLS functionality.

```
CE1#show run
Building configuration...

Current configuration : 1966 bytes
!
! Last configuration change at 13:13:24 UTC Sat Jul 12 2025
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
!
hostname CE1
!
boot-start-marker
boot-end-marker
!
no aaa new-model
no ip icmp rate-limit unreachable
ip cef
!
no ip domain lookup
no ipv6 cef
!
!
multilink bundle-name authenticated
!
ip tcp synwait-time 5
!
!
interface FastEthernet0/0
 ip address 192.169.1.1 255.255.255.0
 duplex full
!
interface Serial1/0
 ip address 192.168.1.2 255.255.255.0
 serial restart-delay 0
```

```
!  
router ospf 1  
  network 192.168.1.0 0.0.0.255 area 0  
  network 192.169.1.0 0.0.0.255 area 0  
!  
ip forward-protocol nd  
!  
control-plane  
!  
!  
line con 0  
  exec-timeout 0 0  
  privilege level 15  
  logging synchronous  
  stopbits 1  
line aux 0  
  exec-timeout 0 0  
  privilege level 15  
  logging synchronous  
  stopbits 1  
line vty 0 4  
  login  
!  
!  
end
```

Appendix 8: Customer Edge Router – CE2

The CE2 router is another Customer Edge router, providing connectivity between the customer's internal network (192.169.2.0/24) and the Pesaro PE router via Serial1/0 (192.168.2.2). Like CE1, it runs OSPF to dynamically advertise and learn routes between the customer site and the provider. It does not use MPLS or VRF, keeping its role focused on simple IP routing and OSPF neighbor formation with the PE. The actual VPN logic and traffic engineering are handled on the provider edge.

```
CE2#show run
Building configuration...

Current configuration : 1904 bytes
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
!
hostname CE2
!
boot-start-marker
boot-end-marker
!
no aaa new-model
no ip icmp rate-limit unreachable
ip cef
!
no ip domain lookup
no ipv6 cef
!
multilink bundle-name authenticated
!
ip tcp synwait-time 5
!
!
interface FastEthernet0/0
 ip address 192.169.2.1 255.255.255.0
 duplex full
!
interface Serial1/0
 ip address 192.168.2.2 255.255.255.0
 serial restart-delay 0
!
!
!
```

```
router ospf 1
 network 192.168.2.0 0.0.0.255 area 0
 network 192.169.2.0 0.0.0.255 area 0
!
ip forward-protocol nd
!
control-plane
!
line con 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
 stopbits 1
line aux 0
 exec-timeout 0 0
 privilege level 15
 logging synchronous
 stopbits 1
line vty 0 4
 login
!
!
end
```

Appendix 9: MPLS Forwarding Tables

Pauillac#show mpls forwarding-table

Local Label	Outgoing Label	Prefix or Tunnel Id	Bytes Label Switched	Outgoing interface	Next Hop
16	16	10.10.10.6/32	2681891	Se1/0	point2point
17	Pop Label	10.10.10.5/32	0	Se1/1	point2point
18	Pop Label	10.10.10.3/32	0	Se1/0	point2point
19	Pop Label	10.10.10.2/32	0	Se1/3	point2point
20	Pop Label	10.1.1.20/30	0	Se1/0	point2point
21	Pop Label	10.1.1.8/30	0	Se1/3	point2point
	Pop Label	10.1.1.8/30	0	Se1/0	point2point
22	Pop Label	10.10.10.4/32	1269061	Se1/2	point2point
23	24	10.10.10.5 0 [5]	0	Se1/0	point2point

Pomerol#show mpls forwarding-table

Local Label	Outgoing Label	Prefix or Tunnel Id	Bytes Label Switched	Outgoing interface	Next Hop
16	Pop Label	10.10.10.6/32	2635287	Se1/2	point2point
17	Pop Label	10.10.10.2/32	0	Se1/1	point2point
18	Pop Label	10.10.10.1/32	650	Se1/0	point2point
19	Pop Label	10.1.1.16/30	0	Se1/0	point2point
20	Pop Label	10.1.1.12/30	0	Se1/0	point2point
21	Pop Label	10.1.1.0/30	0	Se1/0	point2point
	Pop Label	10.1.1.0/30	0	Se1/1	point2point
22	17	10.10.10.5/32	0	Se1/0	point2point
23	22	10.10.10.4/32	1299211	Se1/0	point2point
24	Pop Label	10.10.10.5 0 [5]	0	Se1/2	point2point

Pulligny#show mpls forwarding-table

Local Label	Outgoing Label	Prefix or Tunnel Id	Bytes Label Switched	Outgoing interface	Next Hop
16	16	10.10.10.6/32	0	Se1/1	point2point
17	17	10.10.10.5/32	0	Se1/0	point2point
18	Pop Label	10.10.10.3/32	0	Se1/1	point2point
19	Pop Label	10.10.10.1/32	0	Se1/0	point2point
20	Pop Label	10.1.1.20/30	0	Se1/1	point2point
21	Pop Label	10.1.1.16/30	0	Se1/0	point2point
22	Pop Label	10.1.1.12/30	0	Se1/0	point2point
23	Pop Label	10.1.1.4/30	0	Se1/0	point2point
	Pop Label	10.1.1.4/30	0	Se1/1	point2point
24	22	10.10.10.4/32	0	Se1/0	point2point

Perugia# show mpls forwarding-table

Local Label	Outgoing Label	Prefix or Tunnel Id	Bytes Label Switched	Outgoing interface	Next Hop
16 [T]	Pop Label	10.10.10.6/32	0	Tu0	point2point
17	18	10.10.10.3/32	0	Se1/0	point2point
18	19	10.10.10.2/32	0	Se1/0	point2point
19	Pop Label	10.10.10.1/32	0	Se1/0	point2point
20	20	10.1.1.20/30	0	Se1/0	point2point
21	Pop Label	10.1.1.12/30	0	Se1/0	point2point
22	21	10.1.1.8/30	0	Se1/0	point2point
23	Pop Label	10.1.1.4/30	0	Se1/0	point2point
24	Pop Label	10.1.1.0/30	0	Se1/0	point2point

```
25      22      10.10.10.4/32      0      Se1/0      point2point
```

[T] Forwarding through a LSP tunnel.
View additional labelling info with the 'detail' option

```
Pescara#show mpls forwarding-table
```

Local Label	Outgoing Label	Prefix or Tunnel Id	Bytes Label Switched	Outgoing interface	Next Hop
16	16	10.10.10.6/32	0	Se1/0	point2point
17	17	10.10.10.5/32	0	Se1/0	point2point
18	18	10.10.10.3/32	0	Se1/0	point2point
19	19	10.10.10.2/32	0	Se1/0	point2point
20	Pop Label	10.10.10.1/32	0	Se1/0	point2point
21	20	10.1.1.20/30	0	Se1/0	point2point
22	Pop Label	10.1.1.16/30	0	Se1/0	point2point
23	21	10.1.1.8/30	0	Se1/0	point2point
24	Pop Label	10.1.1.4/30	0	Se1/0	point2point
25	Pop Label	10.1.1.0/30	0	Se1/0	point2point
26	No Label	192.168.1.0/24[V]	\		
			0	aggregate/CUSTOMER1	
28	No Label	192.169.1.0/24[V]	\		
			1207791	Se1/1	point2point

```
Pesaro#show mpls forwarding-table
```

Local Label	Outgoing Label	Prefix or Tunnel Id	Bytes Label Switched	Outgoing interface	Next Hop
16	Pop Label	10.10.10.3/32	0	Se1/0	point2point
17	17	10.10.10.2/32	0	Se1/0	point2point
18	18	10.10.10.1/32	0	Se1/0	point2point
19	19	10.1.1.16/30	0	Se1/0	point2point
20	20	10.1.1.12/30	0	Se1/0	point2point
21	Pop Label	10.1.1.8/30	0	Se1/0	point2point
22	Pop Label	10.1.1.4/30	0	Se1/0	point2point
23	21	10.1.1.0/30	0	Se1/0	point2point
24	22	10.10.10.5/32	0	Se1/0	point2point
25	23	10.10.10.4/32	0	Se1/0	point2point
26	No Label	192.168.2.0/24[V]	\		
			440	aggregate/CUSTOMER1	
28	No Label	192.169.2.0/24[V]	\		
			2555599	Se1/1	point2point